Hydrogen
embrittlement
(see 4.3)
A-7 Double tube structures ensuring no leakage, or fixed hydrogen detectors being capable of detecting a hydrogen leak, should be provided for confined places where leakage of hydrogen may occur, such as cargo valves, flanges and seals.
Susceptibility to
leakage
(see 4.4.2)

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occur, such as cargo valves, flanges and seals.
Susceptibility to
leakage
(see 4.4.2)
A-8 Helium or a mixture of 5% hydrogen and 95% nitrogen should be used as the tightness test medium for cargo tank and cargo piping.
Susceptibility to
leakage (see
4.4.3)
A-9 The amount of carbon dioxide carried for a carbon dioxide fire-extinguishing system should be sufficient to provide a quantity of free gas equal to 75% or more of the gross volume of the cargo compressor and pump rooms in all cases.
Fire by Hydrogen
(see 4.7.3)
Wide range of
flammability limits
(see 4.10)
A-10 When deterioration of insulation capability by single damage is possible, appropriate safety measures should be adopted taking into account the deterioration.
High pressure
(see 4.8)
A-11 Appropriate measures should be provided to prevent vents becoming blocked by accumulations of ice formed from moisture in the air.
Low temperature
(see 4.2.2)

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becoming blocked by accumulations of ice formed from moisture
in the air.
Low temperature
(see 4.2.2)
A-12 Due consideration should be given to means for handling boil-off

gas.
High pressure
(see 4.8)
A-13 Due consideration should be given to static electricity associated with rotating or reciprocating machinery including the installation of conductive machinery belts and precautionary measures incorporated in operating and maintenance procedures , in addition to the bonding of tanks, piping and equipment required by paragraph 5.7.4 of the Code.
Anti-static clothing and footwear, and a portable hydrogen detector should be provided for each crew member working in the cargo area.
Static electricity
(see 4.9.2)
A-14 The cargo operation manuals required in paragraph 18.2 of the Code should include limitations of various operations in relation to environmental conditions.
Wide range of
flammability limits
(see 4.10)

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to environmental conditions.
Wide range of
flammability limits
(see 4.10)
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No. Special requirement Related hazard
A-15 An appropriate procedure should be established for warm -up, inert gas purge, gas -free, hydrogen purge and pre -cooling. The procedure should include:

.1 selection of inert gas in relation to temperature limit;

.2 measurement of gas concentration;

.3 measurement of temperature;

.4 rates of supply of gases;

.5 conditions for commencement, suspension, resuming and termination of each operation;

.6 treatment of return gases; and

.7 discharge of gases.

Prevention of
dangerous
purging operation
(see 4.11)
A-16 Only almost pure para-hydrogen (i.e. more than 95%) should be loaded in order to avoid excessive heating by ortho - to para-hydrogen conversion.
General
(see 4.1)
A-17 Fire detectors for detecting hydrogen fire should be selected,
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para-hydrogen conversion.
General
(see 4.1)
A-17 Fire detectors for detecting hydrogen fire should be selected, taking into account the features of hydrogen fire, to the satisfaction of the Administration.
Features of
hydrogen fire and
fire hazard
(see 4.7.4)
A-18 At the design stage, dispersion of hydrogen from vent outlets should be analysed in order to minimize risk of ingress of flammable gas into accommodation spaces, service spaces, machinery spaces and control stations. Extension of hazardous areas should be considered based on the results of the analysis.
Low density and
high diffusivity
(see 4.5)
A-19 Due consideration should be given to appropriate safety measures to prevent formation of explosive mixture in the case of a leakage and permeation of hydrogen, including:

.1 installation of hydrogen detectors in order to detect a possible ground -level travel of low temperature hydrogen gas, and at high points in spaces where warm
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a possible ground -level travel of low temperature
hydrogen gas, and at high points in spaces where warm
hydrogen gas can be trapped; and

.2 application of "best practice" for land -based liquid hydrogen storage taking into account appropriate guidance such as "Cryogenics Safety Manual Fourth Edition (1998)"8).
General

(see 4.1)
A-20 In the case that fusible elements are used as a means of fire detection required by paragraph 18.10.3.2 of the Code, flame detectors suitable for hydrogen flames should be provided in addition at the same locations. Appropriate means should be adopted to prevent the activation of ESD system owing to false alarm of flame detectors, e.g. avoiding activation of ESD system by single sensor (voting method).
Fire hazard
(see 4.7.4)

No. Special requirement Related hazard
A-21 Consideration should be given to enhance the ventilation
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No. Special requirement Related hazard
A-21 Consideration should be given to enhance the ventilation capacity of the enclosed spaces subject to liquefied hydrogen leakage and permeation, taking into account the latent heat of vaporization, specific heat and the volume of hydrogen gas in relation to temperature and heat capacity of adjacent spaces.
Low density and
high diffusivity
(see 4.5)
A-22 Liquid and gas hydrogen pipes should not pass through enclosed spaces in addition to other than those referred to in paragraph 5.2.2.1.2 of the Code, unless: